
Semantic properties of abstract prompts shape sequential decision making in design

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Abstract

The challenge of design is to take an abstraction to a fully realized, fit-for-purpose creation. Through a complex set of sequential decisions, the designer defines, explores, and refines the space of possibilities, converging on one of many possible solutions. The process often begins with a set of requirements paired with abstract inspiration (e.g., “the design should be *delightful*”). While such semantic information is both common to the design process and known to influence creativity, the mechanism through which abstract, linguistic information shapes design decision making is unknown. Here, we characterize sequential decision making in design and demonstrate how semantics impact designers’ choices using Bayesian modeling approaches. We show that individuals employ different strategies in solving a design challenge, with a majority of participants acting to adjust aesthetic, rather than functional, features in order to realize their intended design. Further, we estimate action trajectories through design space and show that sequential design decisions are influenced by the semantic properties of the design prompt. Both imageability, or the degree to which a concept can be visualized mentally, and number of semantic neighbors are associated with diversity of action patterns, suggesting that prompts that are more readily represented visually and through relationships with other words can drive more diverse behaviors. In addition to providing mechanistic insight into how decisions are made in design tasks, this work identifies computational targets for interventions to support creative problem solving and beyond.

Keywords: design, creativity, sequential decision making, problem solving

*Authors contributed equally to this work.

1 Introduction

Design involves a complex cognitive process that begins with an idea and results in an artifact that meets specific functional and aesthetic criteria. In practice, these criteria are often accompanied by vague, semantic cues (e.g., *magical*, *ultimate*) that designers and other stakeholders hope will lead to uniquely innovative designs [8]. The ‘how’ for realizing such hopes is unknown. Prior work has shown that the semantic properties of design cues are related to the originality and fluency of creative idea generation [2] as well as characteristics of the final design [9]. Nonetheless, the mechanism through which abstract semantic cues influence design decision making is unknown.

Here, we interrogate the process through which the space of possible designs is defined, explored, and refined when trying to meet a design goal. By instrumenting a digital interface comparable to those used by professional designers, we were able to observe the sequential decision making processes as research participants generated novel designs in response to abstract prompts. The semantic properties of prompts varied in two dimensions relating to how language-based cues may impact design decision making: imageability and the number of semantic neighbors. Given a word, **imageability** reflects the ease of generating an associated mental image and may be particularly important for realizing a visual design. The **number of semantic neighbors** reflects how related a word is to other words; words more closely related to other words may be more easily translatable to a design task because of their meanings or likely utility are more accessible. Furthermore, the large action space of the interface not only enhanced the ecological validity of the experiment, but also enabled creative freedom by offering diverse choice options more likely to capture participants’ intentions. We expected that the way in which participants completed the task (i.e., the actions they took to achieve the design goal) would be impacted by the characteristics of the design prompt.

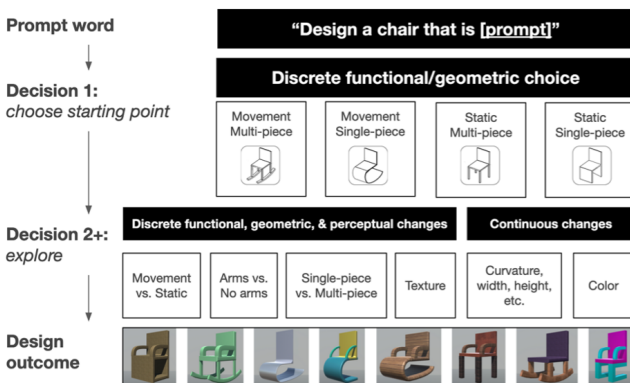
2 Methods

2.1 Design Task

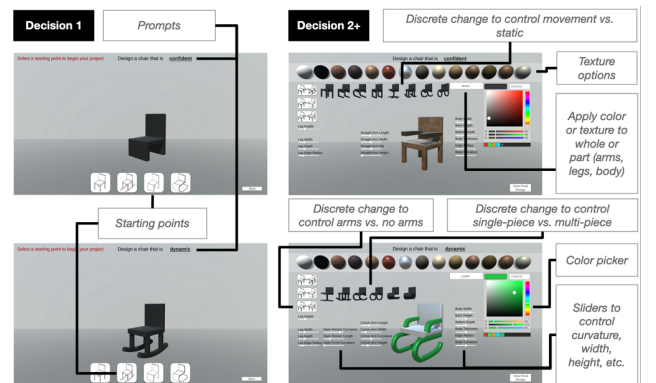
Eligible US-based participants (N=480) were recruited via Prolific and completed a Design Task via Qualtrics. Participants were compensated at a rate of \$15/hr for their time (M = 24.33 min., SD = 16.08 min.). Methods for research with human subjects are described briefly below; further details are in Nandy et al. [9].

Each participant used a custom Unity interface [3, 1] to design a chair that reflected one of 11 specific prompt words: *versatile*, *dependable*, *modern*, *dynamic*, *confident*, *unique*, *exciting*, *sophisticated*, *friendly*, *cheerful*, *graceful*. Prompt words were drawn from a set of semantic descriptors previously associated with product design and varied in two item-level linguistic properties: imageability [7] and number of semantic neighbors [6]. Actions were logged from the interface during the design task at a granular level, corresponding to different interface elements (buttons and sliders).

Participants were instructed to “Design a chair that is [prompt word].” Participants first selected one of four discrete combinations of function and modularity as their starting point: static and multiple pieces (most closely reflecting a “prototypical” four-leg chairs), static and singular piece (a change in geometry), moving and singular piece (a change in function), or moving and multiple pieces (a change in function and geometry). After selecting the starting point, participants advanced to a screen where all other possible actions were available to them. Importantly, participants were able to freely explore their options and make minor or wholesale revisions to their design without any time constraints. Actions included discrete, geometric, or perceptual changes to the design. Discrete changes addressed the chair’s function (able



(a) Design Task structure and example participant designs



(b) Design Task interface

Figure 1: Design Task structure and interface.

to move vs. static, having arms vs. no arms) or modularity (multiple pieces vs. a singular piece). Geometric changes continuously modified the size or shape of the chair. Perceptual changes modified the color or texture of the chair. The full task structure is shown in Figure 1a. Some changes, like color and texture, could be changed for the whole design or for individual parts as shown in Figure 1b. Available choices were not exhaustive but were limited by the set of chairs available in online 3D model repositories (e.g., the chair category of the ShapeNet dataset [4]).

After completing the design task, participants were asked to rate (1-7) and describe their satisfaction with their design, indicating both the extent to which it reflected the prompt ("How well does your final design satisfy the prompt 'design a chair that is [prompt word]'") as well as their personal appraisal ("How satisfied are you with your final design?"). Participants also answered questions about their level of engagement and perceived creativity in addition to providing demographic information about themselves.

2.2 Action Analysis

Design Task actions corresponding to different interface elements (buttons and sliders) were logged and coded to indicate the type of action. Some elements corresponded to the structure of the task and interface, such as the buttons indicating the first decision (i.e., starting style) and buttons that indicated progress. Other elements led to discrete changes (function or modularity, enabled by buttons), geometric changes (size or shape, enabled by sliders), or perceptual changes (color or texture, enabled by buttons, a slider, and a dropdown) in the design.

First, state transition probabilities were estimated in the full dataset (irrespective of prompt word) to establish an expectation on behavior in the Design Task. Each row of the transition matrix represented the probabilities of transitioning from the current state to the next state, given the set of all possible states (each row sums to 1). A Dirichlet distribution was used to estimate probabilities for each row in the transition matrix. A prior was set based on knowledge about transition probabilities influenced by the interface design (e.g., the buttons in the exploration phase could only be reached after completing the starting phase). Then, the actual transition frequencies from the observed data were used to update the prior and obtain the posterior distribution for each row of the transition matrix. Transition probabilities are visualized in Figure 2, for all conditions.

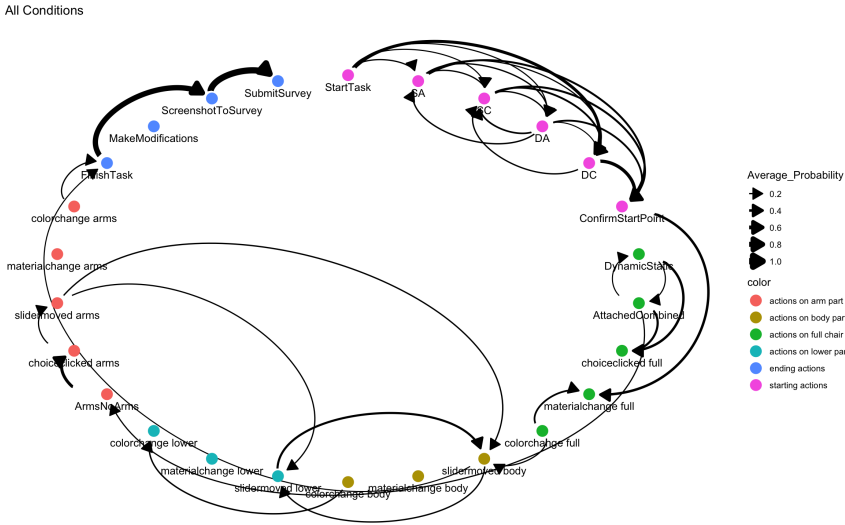


Figure 2: State transition probability diagram. Only transition probabilities over the threshold of 0.15 are visualized.

Levenshtein distance (edits from insertions, deletions, or substitutions) with each action represented by a unique character. Then, the distance between each participant’s action path and the medoid path was used as an index of individual differences in designer behavior.

Participants’ actions were also clustered to provide insight into overall strategies for solving the design challenge. Only actions in the exploration phase (Decision 2+) were considered, as well as only actions that actually changed the design. First, the action sequences were converted to unique strings that accounted for both the action purpose (discrete changes, geometric changes, or perceptual changes) as well as the part of the chair it was applied to (lower, full, body, arms).

In order to test the effect of the prompt word while accounting for interface constraints, a leave-one-out scheme was employed for each condition. The prior was updated with observed data from all of the conditions other than the one left out, resulting in a baseline posterior distribution. Then, this distribution was updated with the left-out data to obtain a final posterior distribution. The KL divergence between the baseline and final posterior distribution of each row was calculated and then added together for a holistic measure of the impact of the left-out prompt word dataset. A larger sum of KL divergences should therefore correspond to a larger change in the row distributions due to the left-out prompt word dataset. This measure of **behavioral diversity** reflects the impact of the prompt word, relative to the other prompts, on the likelihood of subsequent decisions.

For inference across participants, we also computed a normative measure of how participants completed the task, the *medoid* action path. This minimum average distance from all other action paths was computed using the

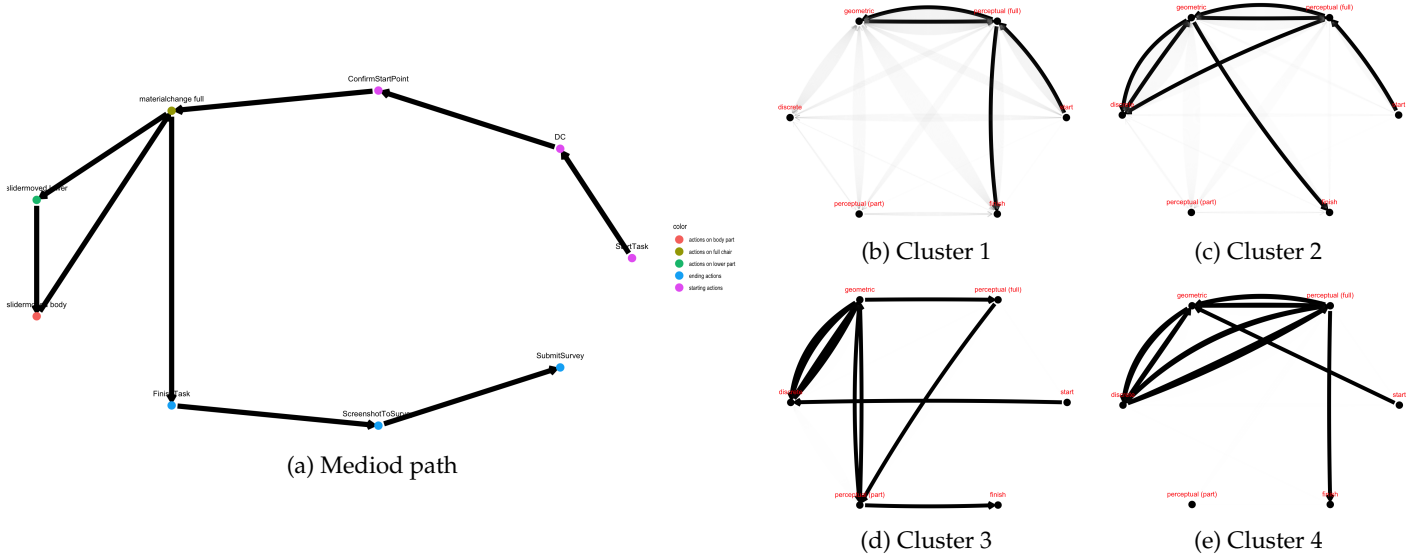


Figure 3: Population-level clustering of action trajectories. Cluster 1, which prioritized actions to change perceptual and geometric features, includes the majority of participants ($\sim 75\%$). Cluster 2 participants ($\sim 24\%$) preempted Cluster 1-type actions with additional actions to adjust discrete features (e.g., changing movement, modularity, or arms). Clusters 3 (6 participants) and 4 (2 participants) include outlier participants who took longer and more varied action paths. Bold lines reflect the cluster mediod while gray lines reflect all action paths.

Then, the Levenshtein distance was utilized as the distance metric. Hierarchical clustering was applied and the number of clusters was determined using the Calinski-Harabasz (C-H) index, resulting in four clusters. The mediod path (path with the minimum average distance to all other paths) is visualized for each cluster in Figure 3.

3 Results

3.1 Behavioral Patterns Vary Across Participants

Figure 3a shows participants’ typical approach to the Design Task (i.e., the mediod path). Hierarchical clustering of action patterns by type revealed population-level phenotypes of design decision making. Given the size of the action space and variation in how participants interpreted the prompt words, we expected that participants would vary widely in how they realized their final designs. As shown in Figure 3, clustering suggests two main approaches to completing the Design Task. The majority of participants ($\sim 75\%$) made only perceptual (color, texture) and geometric (size, shape) changes to their designs, first refining perceptual and then geometric features (Fig. 3b). Most other participants ($\sim 24\%$) first made choices over discrete features (chair functionality) and then alternated between geometric and perceptual features (Fig. 3c). A few remaining participants took alternative paths to create their designs (Figs. 3d and 3e).

3.2 Prompt Words Differentially Impact Action Trajectories

State transition probability matrices reflect the decision trajectory through design space. We expected that higher imageability and number of semantic neighbors would be associated with more diverse action trajectories because a more robust mental representation of the prompt might support wider exploration of design space. Figure 4 shows how the semantic properties of the prompt word impacted sequential decision making in the task. The high imageability word “cheerful” had the largest KL divergence, indicating that there was the greatest impact of changing the transition probabilities from that condition. Words with high imageability and a high number of semantic neighbors tended to have larger effects on participants’ actions than words with both low imageability and a low number of semantic neighbors.

4 Discussion

Design has long operated under the assumption that abstract semantic cues can help guide designers to innovative designs. How words like *modern* and *dynamic* actually influence the process of design is less well understood. We provide one of the only, to our knowledge, computational accounts of empirical decision making processes in design [10] and provide evidence for *how* semantic prompts impact design. These findings are aligned with our previous work, where

we showed the semantic properties of design prompts could predict features of final designs. Prompt word imageability predicted the vibrance of the final design while the number of semantic neighbors was negatively associated with the functional complexity of the design [9].

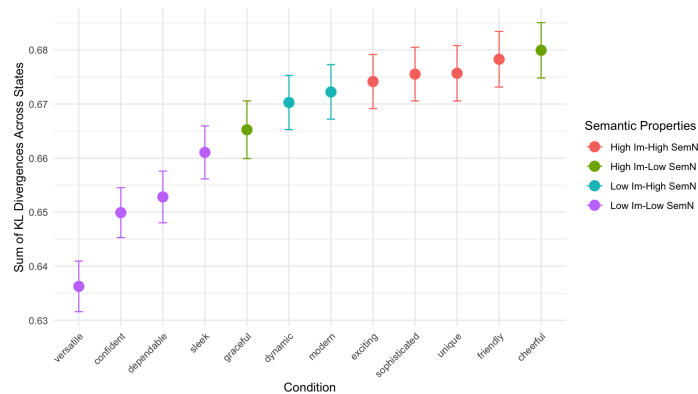


Figure 4: Variance in state transition probabilities by prompt word condition. Higher KL divergence indicates higher between-participant variance in action trajectories.

may have been impacted by the Design Task’s online nature in ways that are difficult to capture (e.g., minimizing actions taken to finish the task or selecting arbitrarily), leading to inaccurate estimation of action tendencies and trajectories.

Despite interface limitations, the Design Task balanced experimental control with relatively high ecological validity, given that design increasingly takes place through digital interfaces. Study results also suggest opportunities for decision and creativity support that can be integrated directly into digital design interfaces, which have a long history of use in professional design, such as Computer-Assisted Design (CAD). By understanding the way semantic information impacts how individuals conceptualize a design task, take actions to realize their vision, and generate a final product; tools can be developed to offer appropriate, timely intervention to encourage exploration of the design space. Interventions can support creativity earlier during ideation or enable more flexible translation between semantic inputs and visual outputs.

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Importantly, these relationships were not deterministic: words with similar semantic characteristics routinely resulted in diverse outcomes [9]. When viewed in light of the results presented here—namely that prompt semantics influence both the types and trajectories of actions—these findings strongly suggest linguistic prompts impact final design features through their effect on sequential decision making. As evidenced by the behavioral phenotypes identified through hierarchical clustering, biases for particular types of choices (i.e., perceptual v. geometric v. discrete) or metacognitive strategies to reduce the complexity of the problem (e.g., “prioritize form over function”) [5] may also mediate the relationship between semantic prompts and the final design. The causal structure of this relationship will be explored in future work.

Future work should also address several limitations. First, in many cases, continuous data were discretized to facilitate model estimation. Further, the task interface did not provide control (via actions) uniformly over the set of all features (e.g., chair geometry), although designer control was prioritized where possible. Finally, some decisions